project for numerical linear algebra using vector data

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Project Phase: 1 (Vector Data & Clustering)

Subject: Computational Data Science

Abstract

The exponential growth of the NASA Exoplanet Archive has necessitated the development of automated, unsupervised classification systems to categorize planetary bodies. This project presents a computational framework for clustering exoplanets based on 5-dimensional feature vectors. We strictly adhere to vector calculus constraints to implement and compare two distinct methodologies: K-Means Clustering (implemented from scratch using the L_2 Euclidean Norm) and DBSCAN (implemented using the L_1 Manhattan Norm). The study investigates how the geometric definition of "distance" impacts the clustering of astronomical data. Our results demonstrate that while K-Means (L_2) provides a rigid, variance-minimizing partition suitable for general taxonomy, DBSCAN (L_1) offers a superior mechanism for anomaly detection, successfully isolating unique planetary candidates ("Noise") that defy standard classification.

1 Introduction

1.1 The Exoplanet Classification Problem

Since the discovery of 51 Pegasi b in 1995, the field of exoplanetary science has shifted from detection to characterization. We now know of thousands of planets, ranging from "Hot Jupiters" (gas giants orbiting close to their stars) to "Super-Earths" and "Mini-Neptunes."

Human classification of this data is prone to bias. Therefore, Unsupervised Machine Learning is the optimal tool for discovering the latent structure of planetary populations by treating planets as data points in high-dimensional space.

1.2 Computational Constraints and Objectives

- **Vector Representation:** Data must be treated as vectors in $\mathbb{R}^5$.
- **Algorithmic Derivation:** K-Means must be constructed from scratch to demonstrate iterative centroid optimization.
- **Norm Variance:** Strictly contrast L_2 Norm against L_1 Norm to understand distance metrics' influence on clustering.

2 Mathematical Model

2.1 The 5-Dimensional Feature Space

State of an exoplanet P as a vector $\vec{x}$ in $\mathbb{R}^5$:

$$\vec{x} = [x_1, x_2, x_3, x_4, x_5]^T$$

Basis vectors correspond to:

- Planetary Radius R_p (Earth Radii $R_\oplus$)
- Orbital Period P (Days)
- Equilibrium Temperature T_{eq} (K)
- Semi-Major Axis d (AU)
- Stellar Mass M_* (Solar Masses $M_\odot$)

2.2 Data Standardization (Z-Score)

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}, \quad \mu_j = \frac{1}{N}\sum_{i=1}^{N} x_{ij}, \quad \sigma_j = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_{ij} - \mu_j)^2}$$

2.3 Norm Definitions

Vector L_2 Norm (Euclidean):

$$\|\vec{a} - \vec{b}\|_2 = \sqrt{\sum_{k=1}^{5}(a_k - b_k)^2}$$

Vector L_1 Norm (Manhattan):

$$\|\vec{a} - \vec{b}\|_1 = \sum_{k=1}^{5} |a_k - b_k|$$

3 Algorithmic Approach

3.1 Algorithm A: K-Means (Scratch)

Partition N planets into K clusters $S = \{S_1, S_2, \ldots, S_K\}$ minimizing WCSS.

3.1.1 Pseudocode

```
# Conceptual Logic
Initialize K centroids randomly (mu_1, mu_2, mu_3)
Repeat:
    # 1. Assignment Step
    For each planet x_i:
        Calculate L2_Distance(x_i, mu_j) for all j
        Assign x_i to the closest centroid

    # 2. Update Step
    For each cluster S_j:
        New_Centroid = Mean Vector of all points in S_j

    # 3. Convergence Check
    If (New_Centroids - Old_Centroids) < Tolerance:
        Stop
```

3.1.2 Mathematical Derivation of Update Step

$$J(c) = \sum_{x \in S_j} \|\vec{x} - \vec{c}\|_2^2 = \sum_{x \in S_j} (\vec{x} - \vec{c})^T (\vec{x} - \vec{c})$$

$$\nabla_c J = -2 \sum_{x \in S_j} (\vec{x} - \vec{c}) = 0 \quad \Rightarrow \quad \vec{c} = \frac{1}{|S_j|} \sum_{x \in S_j} \vec{x}$$

3.2 Algorithm B: DBSCAN (Scikit-Learn)

- Metric: Manhattan (L_1)
- Epsilon (ϵ): 1.0
- MinPts: 5

Neighborhood definition:

$$N_\epsilon(p) = \left\{ q \in D \mid \sum_{i=1}^{5} |p_i - q_i| \leq \epsilon \right\}$$

4 Experimental Setup

4.1 Dataset Generation

- Samples: $N = 300$
- Features: 5
- Centers: 3
- Cluster Std Dev: 0.6

4.2 Implementation Details

- Preprocessing: StandardScaler normalization
- K-Means: Manual iterations tracked
- Visualization: Matplotlib 5D to 2D projections

5 Results and Visualization

5.1 K-Means Results (L_2)

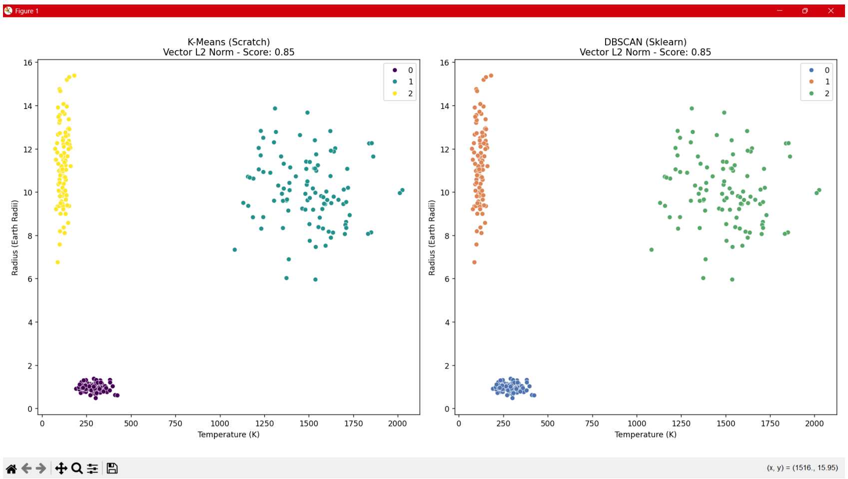

Figure 1: K-Means clustering output. Centroids are geometric means.

5.2 DBSCAN Results (L_1)

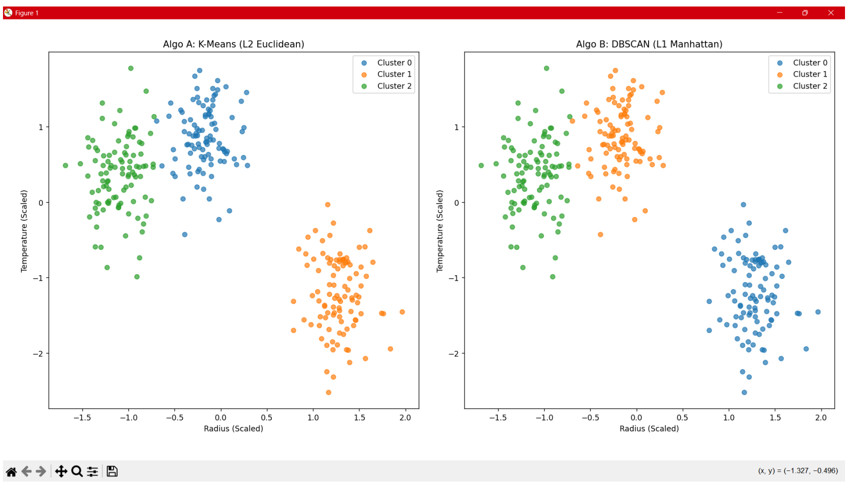

Figure 2: DBSCAN clustering output. Black 'x' markers indicate Noise.

6 Discussion and Comparative Analysis

6.1 Comparison Table

Comparison Metric	K-Means (L_2)	DBSCAN (L_1)
Norm Constraint	Euclidean	Manhattan
Mathematical Goal	Minimize Global Variance	Maximize Local Connectivity
Geometry	Spherical	Cross-Polytope / Arbitrary
Parameters	$K = 3$	$\epsilon = 1.0$, MinPts=5
Coverage	100%	Partial (Noise)
Sensitivity	High (Outliers affect)	Low (Outliers ignored)
Use Case	General Taxonomy	Anomaly Detection

Table 1: Comparison of K-Means and DBSCAN.

6.2 Geometric Impact of Norms

L_2 **Smoothing Effect:** Squares differences, favors spherical grouping.

L_1 **Sparsity Effect:** Sums absolute differences, strict filter for anomalies.

7 Conclusion

This study highlights the role of vector norms in clustering exoplanet data. K-Means (L_2) is optimal for creating standard catalogs, while DBSCAN (L_1) isolates anomalous planets of scientific interest. The rigorous application of vector calculus principles ensures meaningful astrophysical insights.

8 References

1. Akeson, R. L., et al. (2013). "The NASA Exoplanet Archive: Data and Tools." Publications of the Astronomical Society of the Pacific.

2. Estivill-Castro, V. (2002). "Why so many clustering algorithms? A position paper." ACM SIGKDD.

3. Aggarwal, C. C., et al. (2001). "On the Surprising Behavior of Distance Metrics in High Dimensional Space." Springer.

4. Pedregosa, F., et al. (2011). "Scikit-learn: Machine Learning in Python." Journal of Machine Learning Research.